9 Solve the following system of linear equations:

$$\begin{cases} |x| + y = 12 \\ x + |y| = 6 \end{cases}$$

L4 Ext

1. Solve $|x - 2| \leqslant 2x - 10$.

2. Solve $|2x - 4| + |3x - 3| \leqslant 5$.

10 **Solve the following absolute-value inequalities**

(1) $|3x + 1| > 2x$ (

8

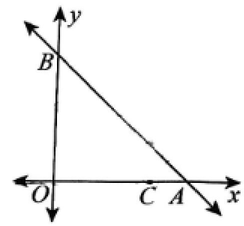
(1) $|2x-3| \geqslant x+1$

⑩ Solve the following absolute-value inequalities:

(1) $|2x + 1| - |5 - x| > 2$

(2) $|x - 1| + |x + 1| < 9$

⑩ The graph of the linear equation $y = -x + 5$ intersects with x-axis and y-axis at A and B respectively .



(3) Points M and N are on the line AB and y-axis respectively, so that the perimeter of $\triangle CMN$ is at its minimum. What are the coordinates of point N?

L8
4

(3) $(2a+b)^2\left[4a^2-(2a-b)b\right]^2 =$ ______

L9 HW
8

(4) $-20x^2 + 9x + 20 =$ ______ .

L9 Extention

5 Factorize: $x^2 + 144y^2 - 25xy =$ ______ .

4 Factorize: $(2x-3y)(3x-2y)+(2y-3x)(2x+3y)-(6x-4y)(6x-9y)=$ ______ .

L10 Extention

7. Given n is a certain natural number in $1 \sim 2021$. If the 72 multiple of n is a perfect square, then the maximum value of n is ______ .

L18 Extention

10 A, B, C, D, E, F form a line, satisfying the conditions that A, B are adjacent, C, D are not adjacent, and E does not stand at the two ends. How many different arrangements are there?

A. 48 B. 96 C. 72 D. 144 E. 288